NLP Problem Statement – ASSIGNMENT PS-3

Title

Intelligent Student Resume Analysis and Internship Recommendation using NLP

Problem Statement

Colleges and universities receive thousands of student resumes for internships and placement activities. Manually analyzing student resumes and recommending suitable internship opportunities is difficult and time-consuming. An intelligent NLP-based system can automate resume analysis, extract relevant skills and qualifications, and recommend suitable internship domains.

In this project, students are required to design and implement an NLP pipeline using a real-world student resume dataset to automatically analyze resumes and classify candidates into suitable internship domains.

Suggested Dataset

• Student Resume Dataset (Kaggle)
• Resume Dataset for Internship Prediction

The developed system should preprocess resume text and perform the following NLP tasks:

1. Neural Language Model (Neural LM):

Develop a Neural Language Model to understand resume content patterns and predict contextual words related to technical domains.

2. Vector Embedding:

Generate vector representations of resume text using Word2Vec, GloVe, or FastText for semantic understanding and profile similarity analysis.

3. POS Tagging:

Implement Part-of-Speech tagging to identify technical skills, educational qualifications, certifications, and project details.

4. Hidden Markov Model (HMM):

Use HMM-based sequence modeling for information extraction and internship domain prediction.

5. Parsing:

Perform syntactic parsing to analyze relationships among skills, projects, internships, and educational background.

The system should classify resumes into internship categories such as:

- Data Science
- Web Development
- Mobile App Development
- Artificial Intelligence
- Cyber Security
- Cloud Computing
- Business Analytics

Final Output

The system should display:

- Extracted student skills
- Recommended internship domain
- Resume category
- NLP analysis results
- Performance metrics

Bonus Marks

Award: 1 Mark Extra Credit will be given for successful implementation using a Virtual Lab.